PART OF nocn GROUP

Functional Skills
Mathematics Assessment
Level 2

There are 2 sections in this assessment, Part A and Part B.



Learner name

_____

Run ID

_____

Learner signature

_____

Centre

_____

Assessment date

_____

Part A - Non-calculator

You have 30 minutes to complete this section.

You must not use a calculator.

asses.

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

64 / 64 Marks (100% Pass · Full Marks)

- Assessment: Sample Paper 1 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [M1/A1]
- Exam Board: NOCN (Ofqual Regulated Functional Skills Reform)
- Pass standard: 38 / 64 marks (Achieved: 64 / 64 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

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NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 1

2456 + (–563) =

Show your working out and write your answer in the box below. **(1 mark)**

Adding a negative number is equivalent to subtraction:

2456 + (-563) = 2456 - 563

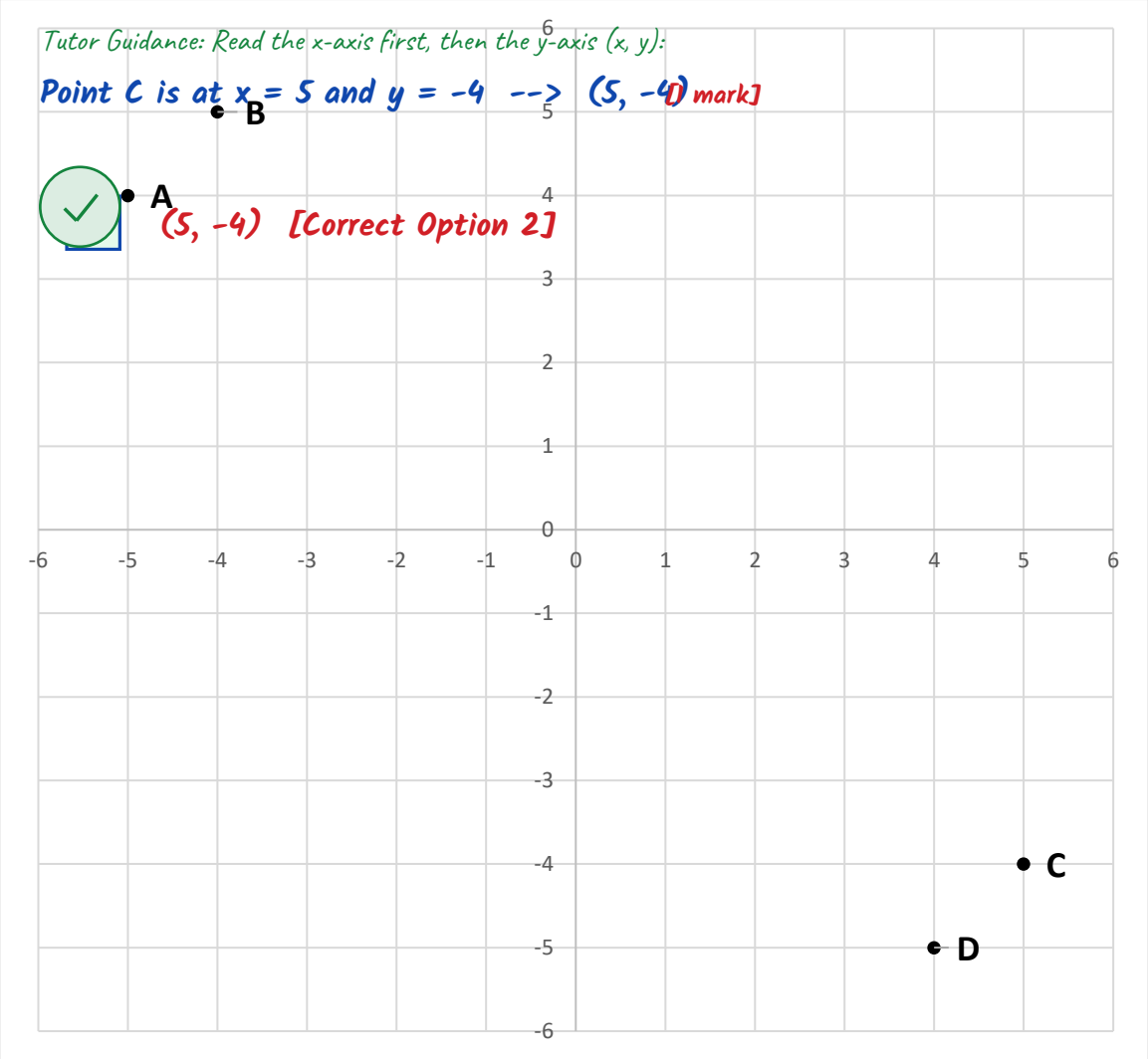
= 1893 [1 mark]

Answer_____

1893 ✓

NOCN Functional Skills Mathematics Level 2 – Reform Sample Paper 1 Assessment Paper

Question 2



What are the coordinates of C? Tick ✓ the correct box. **(1 mark)**

1) **(-4, 5)** ☐

2) **(5, -4)** ☐

3) **(4, -5)** ☐

4) **(-5, 4)** ☐

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 3

Find the mode of this data.

36400g 54.2kg 36.4kg 56kg 22.1kg 4300g 51.5kg 75kg

Step 1: Convert all values to kilograms (1 kg = 1000 g)

- 36400 g ÷ 1000 = 36.4 kg
- 4300 g ÷ 1000 = 4.3 kg

[1 mark for conversion]

Show your working out and write your answer in the box below. **(2 marks)**

Step 2: List all values in kg to identify the mode (most common value):

36.4 kg, 54.2 kg, 36.4 kg, 56 kg, 22.1 kg, 4.3 kg, 51.5 kg, 75 kg

36.4 kg appears twice (all other values appear once).

Mode = 36.4 kg [1 mark]

Answer 36.4 kg (or 36400 g) ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 4

SCALE: 1 square = 0.5 m (or 2 squares = 1 metre) [2 mark scale]

Tim has decided to landscape his garden. His garden is a rectangle 11.5m by 15.75m.

Garden Dimensions: 11.5 m (23 squares) by 15.75 m (31.5 squares)

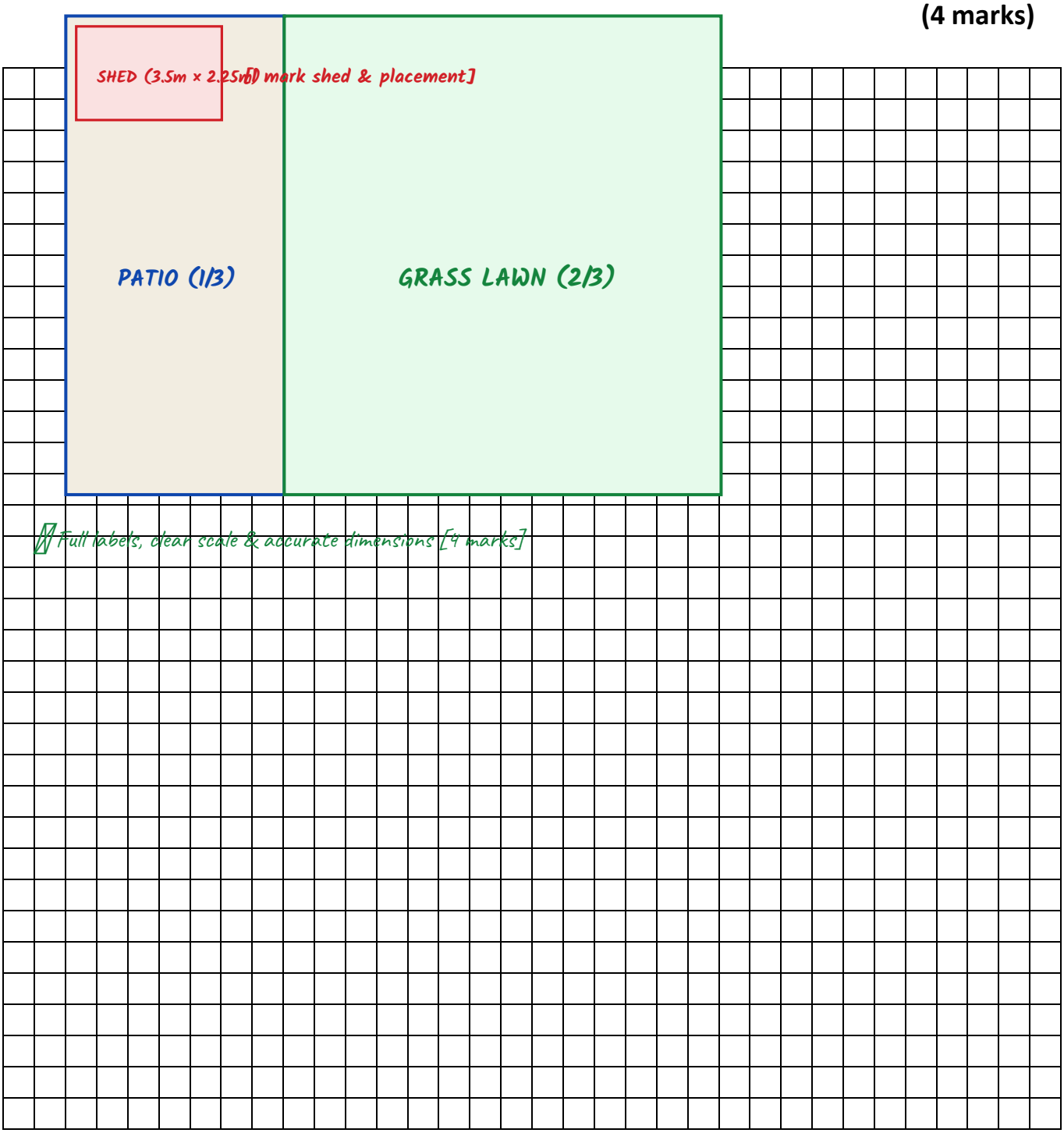
He wants to have two thirds grass and the rest as a patio. He also wants a shed 3.5m x 2.25m to be placed in any corner of the patio. [1 mark grass proportion]

Patio = 1/3 of garden = 11.5 m × 5.25 m (10.5 squares)

Grass = 2/3 of garden = 11.5 m × 10.5 m (21 squares)

Draw a scale drawing on the graph paper below, label the drawing clearly.

(4 marks)



NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 5

Step 1: Find common denominator for 6 and 4 (LCM = 12):

5/6 = (5 x 2) / (6 x 2) = 10 / 12

13/4 = (13 x 3) / (4 x 3) = 39 / 12

[1 mark common denom]

Step 2: Add fractions and convert to mixed number:

Show your working out and write your answer in the box below. **(2 marks)**

10/12 + 39/12 = 49 / 12 = 4 1/12

[1 mark answer]

49 / 12 (or 4 1/12) ✓

Answer_____

Question 6

Step 1: Subtract deposit from total cost of house:

£1,000,000 - £378,500

[1 mark method]

= £621,500 borrowed from bank

[1 mark answer]

Allen is buying a house, it costs 1 million pounds.
He has a deposit of £378,500. The remaining amount he will borrow from the bank.

How much will Allen borrow from the bank?

Show your working out and write your answer in the box below. **(2 marks)**

£621,500 ✓

Answer_____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 7

Calculate the area of a triangle where height = 15cm and base = 7cm.

Formula: Area of triangle = (base × height) ÷ 2

Show your working out and write your answer in the box below. **(2 marks)**

Area = (7 cm × 15 cm) ÷ 2 = 105 ÷ 2 = 52.5 cm² [1 mark method]

52.5 cm² ✓ [1 mark units & value]

Answer

Step 1: Total number of balls = 16 green + 4 purple + 5 blue = 25 balls [1 mark]

Step 2: P(blue) = 5 / 25 = 1 / 5

Convert to percentage: 1/5 × 100% = 20% (or 20.00% to 2dp) [1 mark]

Question 8

What is the probability of picking a blue ball from a box with 16 green, 4 purple and 5 blue? Write your answer as a percentage to 2dp.

Show your working out and write your answer in the box below. **(2 marks)**

20% (20.00%) ✓

Answer_____

End of Part A
Non-calculator

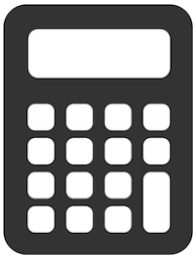
The invigilator will collect this section when the assessment time has elapsed.

The section must be collected by the invigilator before proceeding to Part B.

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Functional Skills
Mathematics Assessment
Level 2



Learner name

_____

Run ID

_____

Learner signature

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Centre

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Assessment date

_____

Part B - Calculator

You have 1 hour 45 minutes to complete this section.

You <u>may</u> use a calculator.

Instructions to learners

Part B: Calculator

You will need a pen, pencil and a ruler. You may need a protractor and compasses.

Use only blue or black ink. Pencil **<u>must only</u>** be used for drawing.

You can write or draw to show your answers.

You have **1 hour 45 minutes** to complete **Part B**.

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 9

Trevor wants to buy a new lorry for work.
Trevor has to transport milk from the farms.



He will need a lorry that is suitable to carry liquid.

The liquid will be carried in a cylinder.

He has seen two lorries for sale:

Lorry 1 has a capacity of 536ft^3

Lorry 2 has the following dimensions: length 19.35ft and diameter 7.6ft

π= 3.14

Trevor wants to buy the larger lorry. Which should he buy? Show clearly how you decided.

Show your working out and write your answer in the box below. **(4 marks)**

Step 1: Calculate radius of Lorry 1 cylindrical tank:

Radius r = diameter ÷ 2 = 7.6 ft ÷ 2 = 3.8 ft *[1 mark radius]*

Step 2: Formula for Volume of Cylinder = π × r² × h (using π = 3.14):

~~Volume =~~ **3.14 × (3.8)² × 19.35** *[1 mark formula]*

Volume = 3.14 × 14.44 × 19.35 = 877.36 ft³ *[1 mark volume]*

Step 3: Compare with Lorry 2 capacity (880 ft³):

Lorry 2 (880 ft³) > Lorry 1 (877.36 ft³). Therefore Trevor should buy Lorry 2 *[1 mark]*

Lorry 2 (capacity 880 ft³ is greater than 877.36 ft ✓

Answer _____ Lorry _____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 10

Dahlia works 21 hours a week.

Her employer says she will be able to increase her hours by 3 in a month's time.

What will be the percentage increase in Dahlia's hours? Round to the nearest whole number.

Step 1: Calculate percentage increase = (Increase ÷ Original) × 100:

Show your working out and write your answer in the box below. **(3 marks)**

Percentage increase = (3 ÷ 21) × 100 [1 mark method]

= 14.2857...% [1 mark value]

Step 2: Round to nearest whole percentage: 14.285...% rounds to 14% [1 mark rounding]

Answer _____

14% ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 11

Terri has read that the span of a person's arm is correlated with their height.

He decides to do an investigation to prove this theory.

He measures the arm span and height of 10 people and records the information in a table.

Person	Arm span cm	Height cm
1	121	125
2	140	136
3	132	140
4	150	150
5	156	164
6	168	162
7	163	170
8	180	175
9	142	140
10	162	159

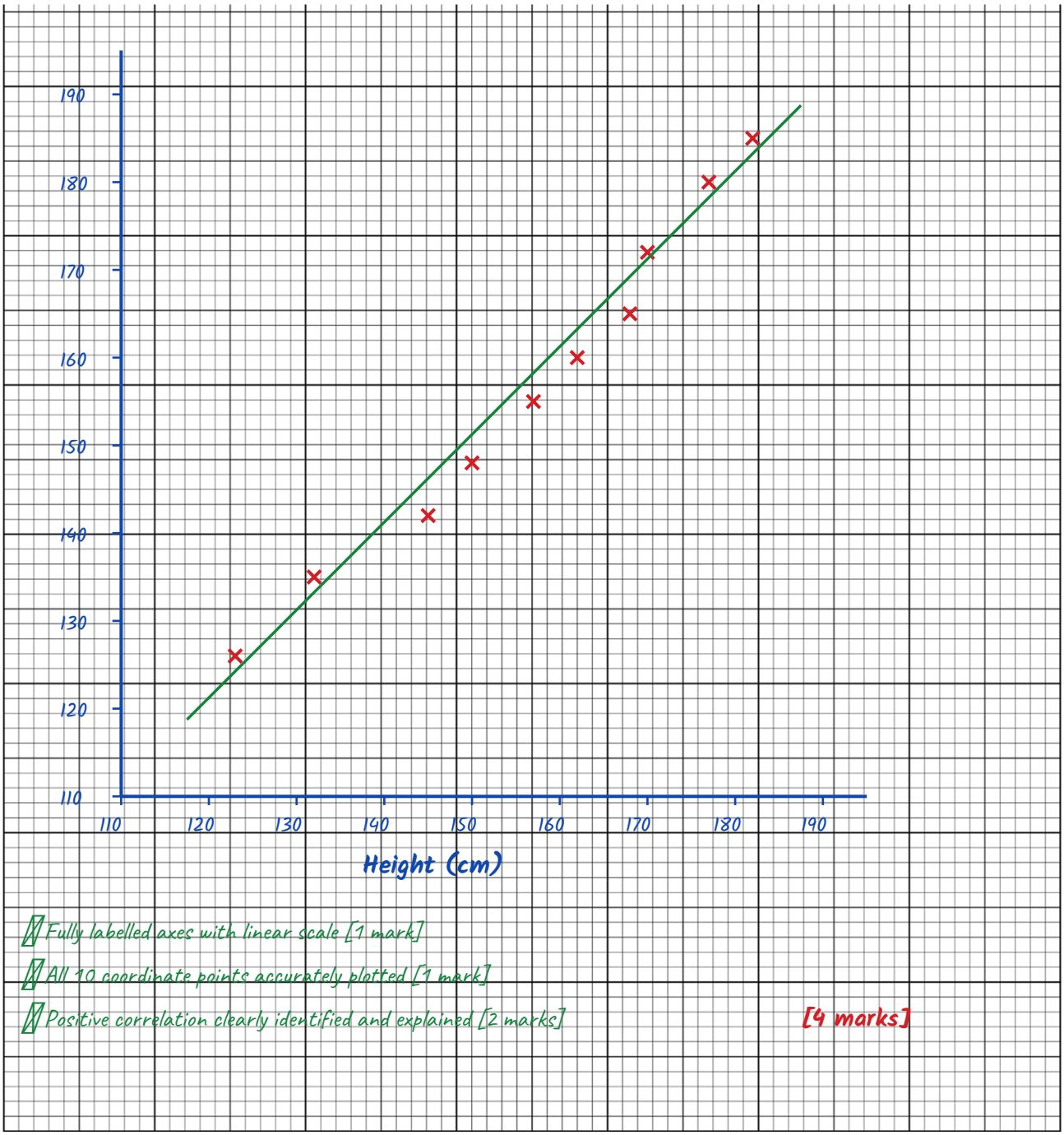
Tutor Guidance: Both variables increase together, showing positive correlation.

Draw a diagram on the next page to show the data supplied.

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

SCATTER GRAPH: Height vs. Arm Span (with Line of Best Fit)

Use this graph paper to draw your diagram. **(3 marks)**



☒ Fully labelled axes with linear scale [1 mark]
☒ All 10 coordinate points accurately plotted [1 mark]
☒ Positive correlation clearly identified and explained [2 marks]

[4 marks]

Describe the correlation the data displays.
Write your answer in the box below. **(1 mark)**

Answer_____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 12

Harry cycles on an indoor track.
The record for distance travelled on an indoor track in an hour is 53.731km.

Harry cycled 41.829km in 47.75 minutes.
Harry thinks if he continued at the same speed he could beat the record.

Could Harry beat the record? Show why you think this.

Show your working out and write your answer in the box below. **(3 marks)**

Step 1: Calculate Harry's speed in km/min:

41.829 km ÷ 47.75 minutes = 0.875999... km/min [2 mark method]

Step 2: Convert speed to km per hour (× 60):

0.876 × 60 = 52.56 km/h [1 mark speed]

Step 3: Compare with record distance in 1 hour (53.731 km):

52.56 km < 53.731 km (short by 1.171 km) [1 mark decision]

Answer_____

No, Harry would not break the record (he covers 52.56 km v… 31 km)

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 13

Dave is organising a charity event for his company. The company will increase the total raised by staff by 23.2%.

Dave has collected £321.45, which is $\frac{3}{5}$ of the money promised by staff.

How much will be raised in total?

Show your working out and write your answer in the box below. **(4 marks)**

Step 1: Calculate amount raised per week:

£321.45 ÷ 3 weeks = £107.15 per week [1 mark]

Step 2: Amount raised over full 5 weeks:

£107.15 × 5 = £535.75 [1 mark]

Step 3: Calculate company 23.2% increase:

£535.75 × 0.232 = £124.294 [1 mark]

Step 4: Total donation = £535.75 + £124.29 = £660.04 [1 mark]

Answer _____

£660.04 ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 14

An Ice Hockey League records the scores for the top two teams after 10 matches.

Team A Goals Scored per match	Team B Goals Scored per match
41	29
42	51
55	32
61	43
27	32
19	57
30	59
44	41
49	33
55	40
Range: 42	Range:
Mean: 42.3	Mean:

Step 1: Calculate Range for Team B (Highest - Lowest):

Range = 59 − 29 = 30 [1 mark]

Step 2: Calculate Mean for Team B (Sum ÷ 10):

(29+51+32+43+32+57+59+41+33+40) ÷ 10 = 417 ÷ 10 = 41.7 [1 mark]

Step 3: Comparative Analysis against Team A (Mean = 44, Range = 35):

- Consistency: Team B has lower range (30 < 35), so Team B was more consistent. [1 mark]
- Scoring Average: Team A has higher mean (44 > 41.7), so Team A scored more. [1 mark]

The coach of Team A says Team A scored more goals on average and were the most consistent goal scorers.

Is the coach correct? Show how you know this.

Team B more consistent (range 30 vs 35); Team A scored higher on average (44 vs 41.7)

Show your working out and write your answer in the box below. **(4 marks)**

Answer_____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 15

Sonia works in a call centre. She wants to calculate the average length of a call.

She records the following information:

Length of call (mins)	Frequency
0 - 2	40
3 - 5	32
6 - 8	30
9 - 11	25
12 - 14	20

Step 1: Calculate midpoints (x) for each interval:

Midpoints: 1, 4, 7, 10, 13 [1 mark midpoints]

Step 2: Multiply midpoints by frequencies (f × x):

1×40=40 | 4×32=128 | 7×30=210 | 10×25=250 | 13×20=260 [1 mark fx values]

Total fx = 40 + 128 + 210 + 250 + 260 = 888 [1 mark total fx]

Total frequency = 40 + 32 + 30 + 25 + 20 = 147 calls [1 mark total f]

Estimated Mean = 888 ÷ 147 = 6.0408... minutes (6.04 mins) [1 mark answer]

Estimate the mean length of a call.

Show your working out and write your answer in the box below. **(5 marks)**

6.04 minutes ✓

Answer_____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 16

Tyler is driving from London to Plymouth.

Tyler travels at an average speed of 40mph covering 32.1 miles to the gallon.

Petrol at his local petrol station is currently 128.5p per litre.

Mileage chart (miles)

3.785 litre to 1 gallon

Step 1: Distance London to Plymouth = 214 miles [1 mark]

Step 2: Fuel needed = 214 ÷ 32.1 = 6.6666... gallons [1 mark]

Step 3: Convert gallons to litres: 6.6666... × 3.785 = 25.233 litres [1 mark]

Step 4: Fuel cost = 25.233 × 128.5p = 3242.48p = £32.42 [1 mark]

Step 5: Travel time = 214 miles ÷ 40 mph = 5.35 hours [1 mark]

Convert 0.35 hours to minutes: 0.35 × 60 = 21 minutes --> 5 hours 21 mins [1 mark]

	Liverpool	London	Manchester	Newcastle	Norwich	Nottingham	Oxford	Penzance	Plymouth	Sheffield	Southend
Liverpool		216	34	167	222	107	168	353	299	74	255
London	216		200	278	114	130	55	282	214	161	43
Manchester	34	200		136	177	68	157	324	281	40	225
Newcastle	167	278	136		252	159	253	481	412	133	299
Norwich	222	114	177	252		118	159	393	327	144	99
Nottingham	107	130	68	159	118		104	323	255	37	160
Oxford	168	55	157	253	159	104		261	193	137	105
Penzance	353	282	324	481	393	323	261		75	361	307
Plymouth	299	214	281	412	327	255	193	75		293	269
Sheffield	74	161	40	133	144	37	137	361	293		197
Southend	255	43	225	299	99	160	105	307	269	197	

Cost: £32.42 | Time: 5 hours 21 mins ✓

How much will the petrol cost and how long will it take to get to Plymouth in hours and minutes?

Show your working out and write your answer in the box below. **(6 marks)**

Answer _____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 17

Alex wants to invest £14000 in a savings account for 2 years.

He finds information about two savings accounts. Both accounts pay interest at the end of each year.

Saver 1	**Saver 2**
3% each year	5.3% for the first year 0.9% for each extra year

Alex thinks he will earn more interest at the end of 2 years with Saver 1.

Is Alex correct? Explain your answer.

Show your working out and write your answer in the box below. **(5 marks)**

Saver 1 (3% compound each year for 2 years):
Year 1: £14,000 × 1.03 = £14,420
Year 2: £14,420 × 1.03 = £14,852.60 (Total Interest = £852.60) [2 marks]

Saver 2 (5.3% year 1, then 0.9% year 2):
Year 1: £14,000 × 1.053 = £14,742
Year 2: £14,742 × 1.009 = £14,874.68 (Total Interest = £874.68) [2 marks]

Comparison: Saver 2 pays £14,874.68 − £14,852.60 = £22.08 more interest.
Therefore Alex is incorrect; Saver 2 is the better investment. [1 mark]

No, Alex is incorrect. Saver 2 pays £22.08 more interest (£87[illegible]s £852.60)

Answer_____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 18

Zeb received a student loan of £12780 last year. Her student accommodation cost £142 per week for 30 weeks.

What fraction of her student loan was used to pay for student accommodation? Write the fraction in its simplest form.

Show your working out and write your answer in the box below. **(2 marks)**

Step 1: Calculate total accommodation cost:

£142 per week × 30 weeks = £4,260

Step 2: Write as a fraction of total student loan (£12,780):

Fraction = 4260 / 12780 [1 mark]

Step 3: Simplify fraction to simplest form (÷ 4260):

4260 ÷ 4260 = 1 and 12780 ÷ 4260 = 3 → 1 / 3 [1 mark]

Answer _____

1 / 3 ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 19

Step 1: Daily rental cost = 8 days × £27 = £216.00

Markus hires a car for 8 days for a work trip. He travels 378 miles. [1 mark substitution]

Step 2: Mileage cost = 378 miles × £0.20 = £75.60

The car hire company use the following formula to calculate the total hire cost:

Step 3: Total cost = £216.00 + £75.80 = £291.60 [1 mark value]

Comparison: £291.60 is £8.40 less than £300 budget. Yes, within budget. [1 mark decision]

Hire Cost = 27d + 0.2m
d = Number of Days
m = Miles Travelled

Yes, £291.60 is £8.40 under the £300 budget ✓

Markus can claim £300 in expenses for car hire.

Will this cover the cost of the car hire? Explain your answer.

Show your working out and write your answer in the box below. **(3 marks)**

Step 1: Division & Multiplication first: (45 ÷ 5) × 3 = 9 × 3 = 27

Step 2: Multiply: 92 × 27 = 2484

Step 3: Add: 2484 + 32 = 2516 [1 mark]

2516 ✓

Answer_____

Question 20

92(45 ÷ 5 x 3) + 32 =

Write your answer in the box below. **(1 mark)**

Answer_____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 21

Rona is a member of a fishing club. She records the weights of the fish she catches in one day.

Fish	Weight(g)
1	4356
2	4851
3	4985
4	4913
5	4803

Step 1: Select median weight from ordered data: 4851 g [1 mark median]

Step 2: Convert 4.851 kg to lbs (1 lb = 0.454 kg):

4.851 ÷ 0.454 = 10.685 lbs (or 10.8 lbs) [1 mark conversion]

The heaviest fish caught by a club member was 10.8lbs.

Step 3: Calculate percentage of total weight:

(0.115 ÷ 10.8) × 100% = 1.064% (or 1.076%) [2 marks]

1lb = 0.454kg

What is the percentage difference between the median weight of fish Rona caught and the heaviest fish caught by a fishing club member?

Show your working out and write your answer in the box below. **(4 marks)**

1.06% (or 1.07%) ✓

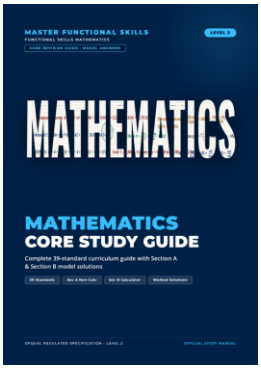
Answer_____

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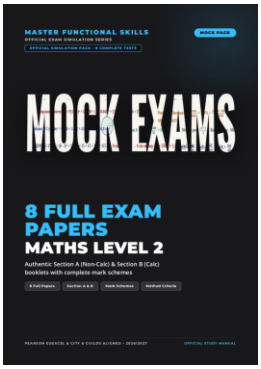


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